Estimation of Fractional Plant Lifeform Cover for the Conterminous United States Using Landsat Imagery and Airborne LiDAR

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***Abstract*— In this study, we produced annual 30-m resolution maps of vegetation lifeform cover (tree, shrub, herb, and an additional class denoted as "other") and their spatially explicit uncertainties for the conterminous United States for the period of 1984–2018, modeled using a quantile regression forest calibrated from a fusion of airborne light detection and ranging (LiDAR)/multispectral data and Landsat imagery. The resulting predictions were compared to available fractional cover products with the same temporal and spatial resolution and to different types of documented land cover change to determine the relative accuracy of the percent cover predictions. We found that our approach yielded a higher accuracy of the shrub, herb, and "other" cover estimates than existing products and had a similar performance of the resulting tree cover estimate compared to other products. Our results indicate that the use of spatially dense training and validation data from airborne multisensor acquisitions facilitates the incorporation of a large range of variability in the response variables into model construction. The produced percent lifeform cover maps, which are publicly available through the Oak Ridge National Laboratory Distributed Active Archive Center, represent a consistent dataset of medium spatial resolution and long temporal record that can be a valuable input into a variety of applications such as the analysis of vegetation change processes, as well as ecosystem conservation and management.**

***Index Terms*— Conterminous United States, Landsat, light detection and ranging (LiDAR), quantile regression forests, spatially explicit uncertainty, vegetation lifeform cover.**

I. Introduction

VEGETATION is an essential component of land–atmosphere interactions, setting controls on the fluxes of water, energy, and carbon dioxide [1], [2], [3]. Thus, information on vegetation's distribution and structure is relevant for understanding and modeling these exchange fluxes and derived effects from vegetation change [4], [5], [6], [7]. Mapping vegetation as a fractional combination of lifeform classes, i.e. tree, shrub, and herbaceous plants, provides a useful framework for identifying spatial and temporal change patterns in vegetation and for modeling atmosphere and biosphere exchanges [8], [9] since plant species that share the same physiognomic aspect are expected to control the fluxes of water, energy, and carbon dioxide in similar ways due to similarities in traits that determine how they use environmental resources [8], [9], [10], [11].

Fractional cover of vegetation, defined as the percentage of an area covered by the vertical projection of plants' leaves, branches, and stems [12], can be derived from remote sensing data and, in recent decades, great efforts have been made to develop continuous fractional vegetation cover datasets across different temporal and spatial scales [13], [14], [15], [16], [17], [18], [19], [20]. These datasets have been produced due to the advancements in remote sensing technology and algorithms, namely, the increasing availability of long time series datasets; improvements in the spectral, spatial, and temporal resolution of the data; the development of machine learning algorithms; and access to high-performance computational infrastructures [21], [22]. However, despite these advances, the construction of vegetation mapping products often remains limited by the availability of training and validation datasets. In general, continuous cover products rely on ground-based field information or photointerpretation of fractional cover of lifeforms at the appropriate scale to construct training and validation datasets. Although ground-based field information and photointerpretation data can be highly accurate, this information is expensive to collect, time-consuming, and labor-intensive, and access constraints to different areas can reduce the availability of data for different ecosystems.

Airborne light detection and ranging (LiDAR) data have become widely used in the assessment of vegetation's vertical and horizontal structure, including the characterization of canopy cover [23], [24], [25], tree height [26], [27], [28], and leaf area index [29], [30], [31]. Different studies show the high accuracy of LiDAR data in describing vegetation structural properties [24], [32], [33], and LiDAR data have become a reliable and accurate measurement of ground-truth vegetation conditions and have been used to evaluate the accuracy of continuous cover products [19], [34]. Considering the increase in the availability and scope of airborne LiDAR data, the high resolution and high accuracy information on vegetation metrics derived from these data and the larger spatial coverage of the airborne LiDAR data compared to field and photointerpretation data; airborne LiDAR data have the

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potential to also be used as the primary calibration dataset, in addition to a validation dataset, for vegetation cover maps construction.

In USA, the acquisition of publicly available airborne LiDAR data has been facilitated by two observation platforms: NASA Goddard's LiDAR, Hyperspectral, and Thermal Imager (G-LiHT) [35] and the National Science Foundation's National Ecological Observatory Network Airborne Observation Platform (NEON AOP) [36]. The available data from these airborne systems sample a large proportion of the different biomes in USA. Therefore, the combined information from the G-LiHT and NEON platforms represents a wide range of vegetation characteristics and composition that would facilitate the incorporation of vegetation heterogeneity into calibration and validation datasets for the construction of fractional cover maps.

For this study, our objective was to produce fractional maps of lifeform cover across the conterminous USA using fused LiDAR/multispectral datasets and Landsat imagery. Three lifeform categories were considered: tree, shrub, and herbaceous ("herb"), as well as an additional category defined as "other" that includes every feature that is either a nonvegetation class or is dead/senescent vegetation. For simplicity, all the cover classes are hereafter referred to as "lifeforms." For each of these categories, maps spanning most of the Landsat archive (1984–2018) were produced, along with a 95% prediction interval to evaluate spatially explicit prediction uncertainty. The resulting fractional cover maps were evaluated: 1) against an independent validation dataset and 2) against other existing fractional cover products.

II. Methods

To achieve the goal of this study, we converted a combined LiDAR/multispectral dataset into 30-m fractional cover estimates of tree, shrub, herb, and "other," and linked them with Landsat surface reflectance (SR) spectra acquired at the same location and within a month of the airborne acquisition date. This combined dataset was used to train and validate quantile regression tree models of per-class fractional cover estimates as a function of Landsat spectra. These models were applied to SR Landsat scenes covering the conterminous USA and for most of the Landsat archive to produce continuous lifeform cover maps. An overview of the model construction workflow is presented in Fig. 1.

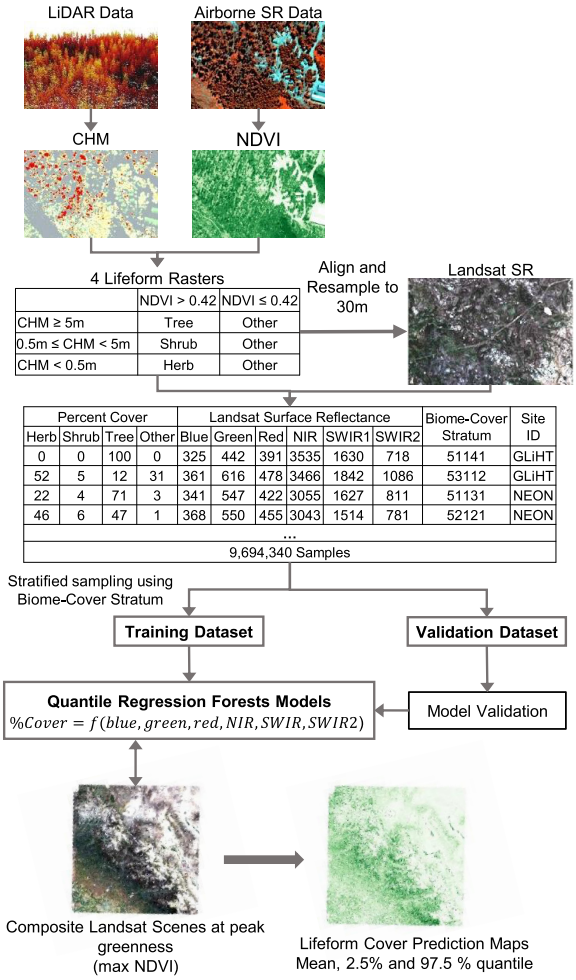


Fig. 1. Lifeform cover models construction workflow. Preprocessing of multisensor airborne data to construct CHM and NDVI. Construction of lifeform cover layers for each acquisition site using NDVI and height thresholds. Unification of the LiDAR/multispectral percent cover and Landsat SR data of all acquisitions in a single dataset and construction of the biome-cover stratum by concatenating the number of the terrestrial biome corresponding to each acquisition and the numbers resulting from classifying each LPC into quantile classes (i.e., 0%–25% = 1, 25%–50% = 2, 50%–75% = 3, and 75%–100% = 4). Separation of training and validation datasets, and construction and application of the quantile regression forest models.

A. Input Data

1) LiDAR/Multispectral Datasets: We used a total of 165 airborne LiDAR datasets acquired during the period of 2011–2017 that cover eight of the 11 terrestrial biomes of the conterminous USA and Alaska, as defined in [37] (Fig. 2), and these biomes represent approximately 86% of the area of the conterminous USA and Alaska. Eighty six of the acquisitions are part of the G-LiHT [35] airborne imager data collection and 79 sites were acquired by the NEON AOP [36]. Acquisitions were selected based on the availability of coupled airborne SR data for each G-LiHT and NEON dataset. We describe the G-LiHT and NEON AOP sensor and collection characteristics in Table I. Acquisitions covering terrestrial biomes in Alaska were included in this analysis to guarantee that the produced lifeform models were able to predict vegetation cover conditions in this region, as prediction maps for this state are planned to be constructed as part of future versions of the lifeform cover product.

The available LiDAR raw point clouds were processed using LAStools [38] and following a standardized workflow based on [39] to guarantee the consistency of the resulting dataset. The raw LiDAR flight lines were tiled into 1000 m^2 with 40-m buffers to avoid edge artifacts. Noise and duplicate points were removed, and points were classified into ground and no-ground returns using a step value of 10 m. Height normalization was applied to each tile and points below 0 m or above 100 m were removed to eliminate outliers. The no-ground points were classified into vegetation and buildings using the lasclassify module, and only vegetation points were retained in the tiles. Finally, using the las2dem module,

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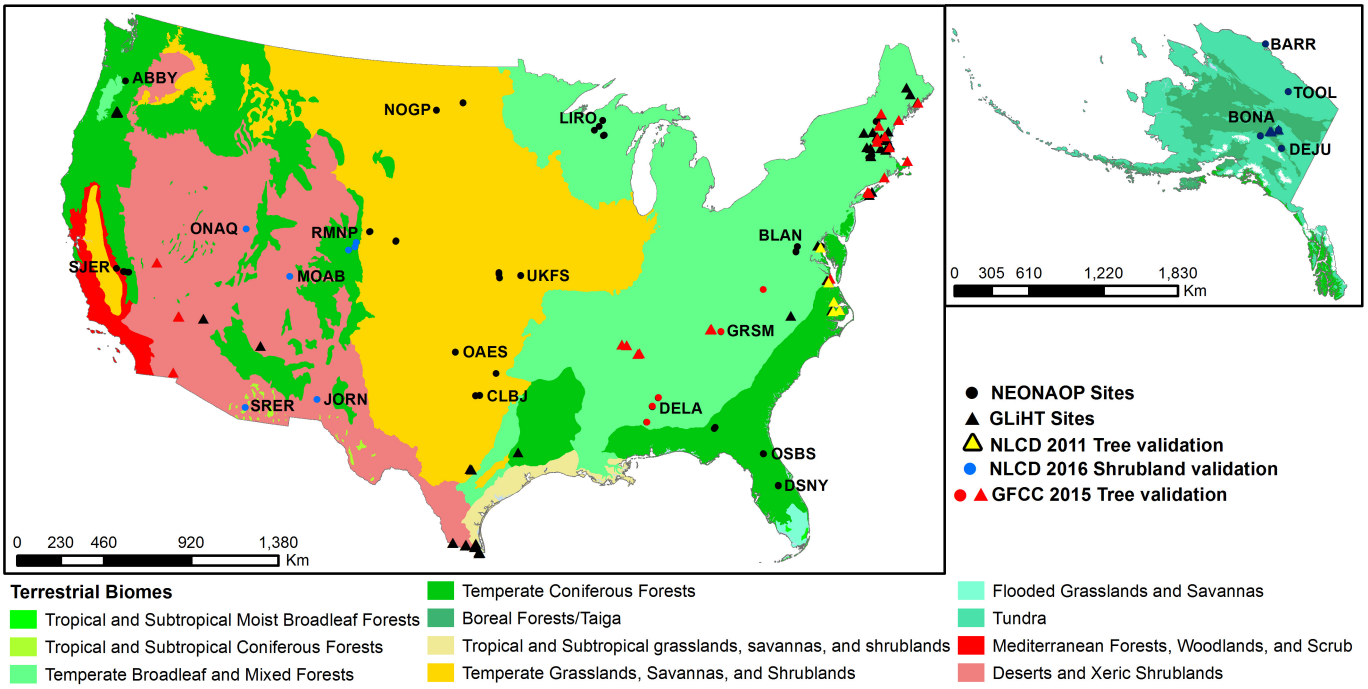


Fig. 2. Airborne LiDAR and multispectral acquisitions across the different terrestrial biomes of the conterminous USA and Alaska (modified from [37]). The acquisitions used for the validation of the different continuous cover products are included in the figure, and the NEON sites used for the NDVI threshold selection and validation are labeled in the figure.

TABLE I
G-LiHT AND NEON AOP SENSOR DESCRIPTIONS AND SPECIFICATIONS

Instrument	Characteristics	G-LiHT	NEON AOP
LiDAR	*Model*	Riegl VQ-480	Optech ALTM Gemini
	Laser wavelength	1550 nm	1064 nm
	Number of returns	up to 8 returns per pulse	up to 4 returns per pulse
	nominal operating altitude	335 m	1000 m
	Laser Beam Divergence	0.3 mrad	0.8 mrad
	Footprint	~ 10 cm	0.8 m
	Point Density	6 points per m2	between 1 and 4 points per m2
Spectrometer	*Spectral range*	400–1,000 nm	380-2510 nm
	Number of bands	402	428
	Spectral resolution	up to 5 nm	5 nm
	Atmospheric correction	Downwelling radiometer	Upward-looking solar spectrometer
Camera	*Spectral range*	-	400 - 700 nm
	Field of view		44 degrees
	Ground sampling distance		0.11 m
	Flight altitude		1000 m

Sensor information from [35], [68], [69]

a 1-m resolution canopy height model (CHM) was constructed, using the first returns of the normalized vegetation points.

The coupled airborne SR data of each G-LiHT and NEON dataset were used to construct 1-m spatial resolution images of normalized difference vegetation index (NDVI) [40].

2) Landsat Imagery: Landsat Collection-1 SR images [Landsat 8 Operational Land Imager (OLI)/Thermal Infrared Sensor (TIRS), Landsat 7 Enhanced Thematic Mapper Plus (ETM+), and Landsat 4-5 Thematic Mapper (TM)] were downloaded from the U.S. Geological Survey (USGS) Earth Resources Observation and Science Center (EROS) (https://www.usgs.gov/centers/eros/) and were used in two steps of the process: for model construction and for the creation of the lifeform cover prediction maps. For both steps, only the visible (V) bands (blue, red, and green), near infrared (NIR), and the two short-wave infrared (SWIR1 and SWIR2) bands were used in the analysis.

For lifeform cover model construction, Landsat scenes overlapping the LiDAR/multispectral acquisitions with the least possible cloud cover and within a month of the airborne acquisition date were selected and downloaded. Each Landsat scene was masked to remove nonclear pixels (pixels with cloud, water, ice, snow, or cloud shadow cover) and cropped using the LiDAR/multispectral acquisition's bounding box.

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To create the lifeform cover prediction maps, we used Landsat SR composite images as inputs. To construct the composite images, all Landsat scenes available from May to August were downloaded from the USGS EROS portal for each path-row scene covering the conterminous USA and for the years 1984–2018. These months were selected because they span most of the growing seasons of different ecosystems in USA, allowing us to bracket the maximum cover signal for the vegetation lifeforms, especially for herb cover. The lifeform cover prediction maps were produced starting in 1984 because previous years of the Landsat TM archive have large areas of CONUS missing.

To create the composite images, all the available images for each path-row-year unique scene were masked for nonclear pixels using the quality assessment layer included in the Landsat file. The first image in the sequence was selected as the reference, and all the other images were aligned and cropped to this reference. NDVI layers were constructed for each of the images in the path-row-year scene, and the date of the maximum NDVI value across layers was identified for each of the pixels. The value for each pixel of the final composite bands corresponded to the pixel's spectral values from the image date with the highest NDVI value [41]. This process allowed us to create images with the least amount of masked areas and atmospheric noise and with the values of reflectance more closely related to peak productivity periods across the growing season for various ecosystems.

B. Training and Validation Dataset Construction

To train and validate the fractional cover models, we constructed a dataset with a 30-m spatial resolution, which included: 1) four response variables derived from the LiDAR/multispectral data—% trees, % shrub, % herb, and % others; 2) six predictor variables derived from the Landsat SR data—Blue, Red, Green, NIR, Short-wave Infrared 1, and Short-wave Infrared 2; and 3) a categorical variable defined as the biome-cover stratum used for separating the data into training and validation: this variable was constructed by concatenating the number of the terrestrial biome corresponding to each acquisition and the numbers resulting from classifying each lifeform percent cover (LPC) into quantile classes (i.e., 0%–25%, 25%–50%, 50%–75%, and 75%–100%).

We based the differentiation of the lifeform types at a 1-m spatial resolution on two sources of information: 1) the NDVI images derived from airborne SR data to separate live vegetation from nonvegetation or dead/senescent-vegetation features (hereafter referred to as "nonvegetation") and 2) the vertical structure of the vegetation derived from the LiDAR data to separate different vegetation height classes.

Since LiDAR data alone are largely insufficient to inform whether a feature is living or nonvegetation, we used the NDVI images from each acquisition to differentiate live vegetation from nonvegetation using a simple threshold value. To calibrate and validate the NDVI threshold, we used NEON acquisitions with available orthophotos. Whenever available, two NEON sites were randomly selected from each of the terrestrial biomes covered by the NEON data in order to have an equal representation of biomes with different characteristics and to guarantee that the selected threshold was adequate to represent differences in live vegetation and nonvegetation features across a wide range of vegetation conditions (Fig. 2). G-LiHT data were not used for this process as the datasets' orthophotos were not available at the time of the analysis.

The orthophotos of each selected site were mosaicked, and a shapefile of 100 random points was created for each acquisition. Each point was photointerpreted and classified into live vegetation and nonvegetation. The NDVI value for each point was extracted from the corresponding NDVI raster for each year and acquisition site, and the resulting information from all the points across all selected acquisitions was consolidated in two datasets corresponding to the threshold selection sites (897 points) and the validation sites (607 points). Using the threshold selection dataset, a beta distribution was fit to the NDVI data of the live vegetation features and a truncated lognormal distribution to the nonvegetation feature class. The 10% quantile of the live vegetation distribution and 90% quantile of the nonvegetation distribution were calculated to determine the final NDVI threshold value of 0.42. The data from the validation sites were used to evaluate the accuracy of the NDVI threshold, which resulted in an overall accuracy of 93% and a Kappa coefficient of 0.85. The NDVI threshold of 0.42 was used to classify the 1-m spatial resolution NDVI raster of each of the 165 airborne acquisitions into live vegetation or nonvegetation.

We defined height thresholds for separating tree, shrub, and herb lifeforms based on an evaluation of different published classification systems and previous works on construction of continuous fields maps [15], [19], [20], [42], [43], [44], [45]. Based on these references, we classified the 1-m spatial resolution CHM into three separate layers: 1) features with less than 0.5 m in height (which would include herb); 2) features equal or above 0.5 m and less than 5 m high (including shrub); and 3) features equal or above 5 m high (including tree).

The classification of each acquisition site into tree, shrub, herb, and "other" was done by merging the classified NDVI and CHM layers such that "other" corresponded to all the features that are nonvegetation, regardless of their height, while herb, shrub, and tree correspond to live vegetation features smaller than 0.5 m, taller or equal to 0.5 m and smaller than 5 m, and greater than 5 m, respectively.

The resulting 1-m spatial resolution lifeform rasters were resampled using pixel averaging to a 30-m pixel resolution and paired and aligned to their corresponding Landsat SR scene. The resampled percent cover rasters were stacked with the cropped Landsat SR bands and transformed into a database (Fig. 1). Those Landsat pixels that were not completely covered by the lifeform rasters, commonly occurring at the edge of the LiDAR/multispectral acquisition areas, were removed from the dataset.

The 30-m spatial resolution data from all the available acquisitions were merged into a single dataset, resulting in a combined dataset that contained $N = 9\,990\,526$ pixels. The biome-cover stratum was used to separate the training and validation datasets to guarantee that all possible combinations of biome and lifeform cover classes were represented in the

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training of the model. The approach to sample from the biome-cover stratum is explained in Section II-C.

C. Lifeform Quantile Regression Tree Models

Using the training data, we constructed a quantile regression forest model for each of the lifeform cover types using the R [46] package quantregForest [47]. Quantile regression forests are a generalization of random forests, which is a machine learning, nonparametric regression method that creates an ensemble of trees by randomly selecting a subset of the training data and a subset of predictor variables to grow each tree and retains the other fraction of the data for error estimation, known as out-of-bag (OOB) error [48]. This injection of randomness results in the reduction of correlation between predictor variables, an increase in prediction accuracy, and a model that is robust to overfitting [48], [49]. In contrast to random forests, which infers the conditional mean of the response variable, quantile regression forests estimate the conditional quantiles, which present more information on the distribution of the response variable, and the dispersion of predicted values for a specific set of independent variables [50]. Using quantile regression forests, we were able to estimate a prediction uncertainty metric by calculating different quantile values of the prediction distribution [51].

The regression models were constructed using a parallel implementation of the quantregForest package with the number of trees to grow (ntree) set to 500 and the number of variables for tree node splitting (mtry) set to two. We did a preliminary data analysis to determine the size of the training dataset that would balance the OOB error reduction and optimization of computational time. For this analysis, we constructed regression models using different values of sample size per biome-cover stratum and evaluated the OOB error of each model, which led to a final sample value of 15 000 for each of the 34 unique biome-cover stratum available in the LiDAR/multispectral dataset, for a total of 510 020 samples for the training dataset. If a specific biome-cover stratum did not have the number of samples required for the training dataset, the available data were sampled with replacement. This way we guaranteed that the training dataset had a balanced representation of each biome-cover stratum for model construction. The resulting training dataset included a total of 203 853 unique samples corresponding to 2% of the 9 990 526 pixels in the complete dataset. All the data samples that were not used in the training subset were included in the validation subset.

D. Model Validation

We evaluated the prediction performance of the resulting LPC models based on the Pearson correlation coefficient (R), root-mean-square error (RMSE), and mean bias error (MBE), using the observed and mean cover prediction values from the validation dataset. Additional evaluation of our models' performance included the comparison of our LPC models against available continuous data products (Table II) and the estimation of fractional cover change for study sites with documented land cover change.

The continuous data products used in the model comparison were the Global Forest Cover Change (GFCC) tree cover product for 2015 [19], and the National Land Cover Database (NLCD) 2011 tree cover [52], [53], [54] and 2016 Shrubland cover products [20]. We chose these datasets because their temporal resolution overlaps with the available dates of the validation data, and they have a spatial resolution of 30 m and present information on continuous variables similar to the ones defined for the LPC models allowing for product comparison.

The predicted cover values of our LPC, the GFCC, and the NLCD products were evaluated against our validation data samples overlapping each product coverage and for the corresponding year. In the case of the NLCD shrubland product, there were no 2016 acquisitions in the validation dataset overlapping the area of the product. However, since overlapping 2017 acquisitions were available, these were used instead for the comparison analysis, and to guarantee equal conditions, 2016 LPC prediction values were used as well.

The analysis of study sites with documented land cover change was done by calculating differences in LPC predicted percent cover from different years to determine whether the lifeform maps were able to track these transitions. For each lifeform cover class, an area of interest was identified and two prediction maps corresponding to the initial and final year encompassing the period of change were selected. The initial year was subtracted from the last year and the resulting differences in percent cover were compared with the land cover change process defined in the literature for each area of interest.

E. Prediction and Uncertainty Map Construction for the Conterminous USA

To produce the final percent cover prediction maps, each quantile regression forest model was applied to the composite images using three prediction arguments: mean, 2.5% quantile, and 97.5% quantile. A prediction uncertainty metric was calculated for each pixel, by correcting the prediction interval between the 2.5% and 97.5% quantiles using a value related to the standard deviation of the binomial distribution

$$\frac{P_{97.5} - P_{2.5}}{\sqrt{P_{\text{mean}} * (1 - P_{\text{mean}})}} \tag{1}$$

where $P_{97.5}$ and $P_{2.5}$ are the 97.5% and 2.5% quantiles, respectively, and P_{mean} is the mean prediction. This correction was established to account for the variation of the prediction interval's width about the prediction mean. Bounded estimates, such as in this case between 0% and 100%, will have a lower variance around the mean for predictions closer to the boundaries due to the predictions not taking values below 0% or over 100%. Conversely, predictions in intermediate values typically have a larger variance because the predictions can take values below or above the mean. This results in smaller prediction intervals around the boundaries and larger intervals in intermediate values, which could mislead into believing that intermediate predictions have higher uncertainty compared to predictions closer to the boundaries. To factor out this variation, the width of the prediction interval is scaled relative to the expected binomial standard deviation.

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TABLE II
CONTINUOUS DATA PRODUCTS SPECIFICATIONS

Data product	Spatial resolution	Temporal resolution	Extent	Response variables	Explanatory variables	Spatially explicit uncertainty	Models constructed
LPC	30m	annual	USA	%tree cover %shrub cover %herb cover %other cover Estimated from airborne LiDAR and multispectral data (resampled to 30 m resolution)	Landsat SR	Prediction uncertainty metric	4 models, one for each lifeform type
GFCC 2015 Tree cover	30m	Every 5 years	Global	%tree cover Estimated from the MODerate-resolution Imaging Spectroradiometer (MODIS) Vegetation Continuous Field (VCF) (250 m spatial resolution)	Landsat SR and Temperature	Root mean squared error	Local regression models were fitted for each Landsat scene in the World Reference System 2, for each epoch evaluated.
NLCD 2011 Tree cover	30m	2011 only	USA	%tree cover Estimated from photointerpretation of 1 m pixel National Agriculture Imagery Program (NAIP) aerial images (resampled to 30 m resolution)	Landsat data and indices, digital elevation models and derivates, previous NLCD data	Standard error	Random forest models were fitted to each NLCD mapping zone
NLCD 2016 Shrubland products	30m	2016 only	USA	% annual herbaceous %bare ground %big sage %herbaceous %litter %sagebrush %shrub Estimated using field data as the training dataset, to fit regression tree models to 2m spatial resolution WorldView-2 images (resampled to 30 m resolution)	Landsat data, ancillary data including slope, aspect, and position index.	Not Available	A regression tree model was fitted for each response variable, for each of three different climatic zones.

The resulting percent cover images for each lifeform and year were mosaicked to create a seamless map. In the areas with overlapping data, pixel values were selected based on the prediction uncertainty so that the pixel values with the lowest uncertainty were retained in the mosaic. As an additional analysis of model performance, we calculated the mean and standard deviation of the prediction uncertainty metric per biome for the lifeform mosaics of 1986 and 2018, to evaluate whether higher prediction uncertainty could be related to deficiencies in the training dataset for specific biomes. We chose these years as a representation of the beginning and end dates of the produced lifeform maps, and 1986 was selected instead of 1984 or 1985 due to these maps having large areas with missing data. The computational workflow and initial data storage were completed in a high-performance computing environment, and to facilitate data access and storage, the final lifeform mosaics are available to the public through the Oak Ridge National Laboratory Distributed Active Archive Center (ORNL DAAC) (https://doi.org/10.3334/ORNLDAAC/1809) [55].

III. RESULTS

A. Model Statistics and Lifeform Cover Prediction Accuracy

The performance metrics of the LPC models, summarized in Table III, resulted in a high correlation between predicted values and the validation data. Overall, the "other" model had the best performance as indicated by the high correlation, and low error values ($R = 0.89$, RMSE $= 16.33\%$, and MBE $= -4.32\%$), and the shrub model had the lowest performance compared to the different lifeform categories ($R = 0.41$, RMSE $= 17.95\%$, and MBE $= -12.21\%$). The predicted percent cover for each model showed a generally positive correspondence with the validation values, as shown in Fig. 3. However, all models tend to overestimate cover values in low cover areas and underestimate percent cover values in high cover areas, which is reflected in the RMSE and MBE values.

TABLE III
MODEL PERFORMANCE METRICS OF THE LIFEFORM COVER MODELS FOR THE COMPLETE VALIDATION DATASET

Lifeform Model	RMSE	MBE	R
Tree	29.46%	-10.14%	0.779
Shrub	17.95%	-12.21%	0.413
Herb	28.72%	-5.40%	0.728
Other	16.33%	-4.32%	0.887

The prediction uncertainty per biome (Table IV) indicates that tree and shrub cover predictions have the lowest uncertainty in the deserts and xeric shrublands biome, while the herb and "other" cover have the lowest uncertainty in the tropical

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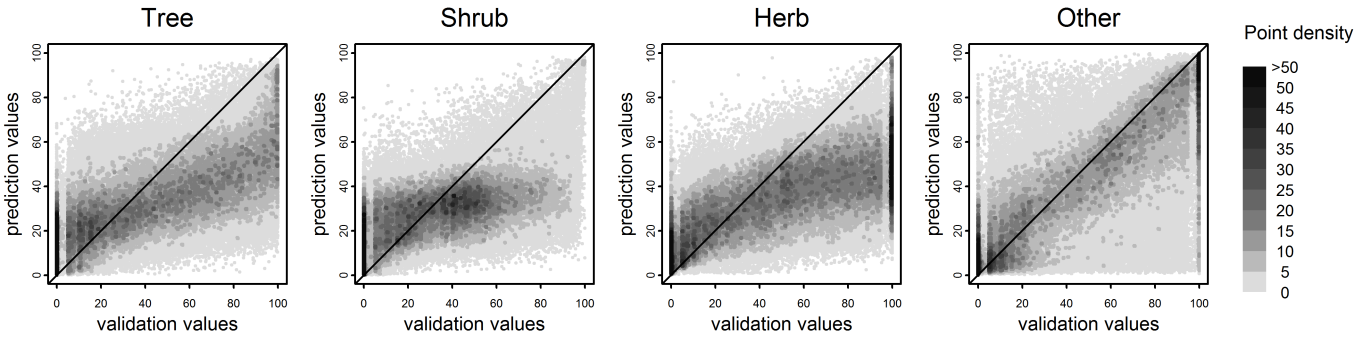


Fig. 3. Scatterplots of predicted versus observed cover values for each lifeform type using a subsample of the validation dataset. The 1:1 line is shown in black for reference.

TABLE IV
MEAN AND STANDARD DEVIATION PER BIOME OF THE PREDICTION UNCERTAINTY FOR THE RESULTING MOSAICS OF 1986 AND 2018

Biome name	Tree cover				Shrub cover				Herb cover				Other cover			
	1986		2018		1986		2018		1986		2018		1986		2018	
	mean	sd	mean	sd	mean	sd	mean	sd	mean	sd	mean	sd	mean	sd	mean	sd
Flooded Grasslands and Savannas	1.63	0.27	1.63	0.26	1.69	0.31	1.60	0.29	1.82	0.30	1.78	0.26	1.88	0.30	1.89	0.43
Tropical and Subtropical Moist Broadleaf Forests	1.55	0.26	1.57	0.28	1.61	0.32	1.57	0.32	1.75	0.29	1.74	0.28	1.82	0.31	1.77	0.37
Deserts and Xeric Shrublands	1.24	0.44	1.22	0.49	1.12	0.43	1.09	0.42	1.65	0.51	1.69	0.53	1.64	0.43	1.65	0.47
Temperate Coniferous Forests	1.58	0.30	1.59	0.33	1.51	0.33	1.47	0.33	1.67	0.30	1.61	0.30	1.70	0.41	1.70	0.52
Temperate Broadleaf and Mixed Forests	1.75	0.29	1.84	0.28	1.76	0.27	1.72	0.25	1.80	0.27	1.72	0.29	1.92	0.44	1.98	0.53
Tropical and subtropical grasslands, savannas, and shrublands	1.59	0.29	1.64	0.28	1.72	0.38	1.67	0.34	1.86	0.26	1.84	0.24	1.87	0.35	1.95	0.45
Mediterranean Forests, Woodlands, and Scrub	1.47	0.29	1.39	0.30	1.36	0.30	1.27	0.33	1.64	0.29	1.58	0.31	1.56	0.33	1.49	0.36
Tropical and Subtropical Coniferous Forests	1.42	0.25	1.40	0.30	1.31	0.24	1.22	0.25	1.49	0.24	1.51	0.30	1.47	0.28	1.42	0.31
Temperate Grasslands, Savannas, and Shrublands	1.51	0.33	1.57	0.36	1.48	0.40	1.47	0.37	1.73	0.29	1.75	0.29	1.73	0.39	1.76	0.45

and subtropical coniferous forests biome. Overall, the highest prediction uncertainty occurs in the temperate broadleaf and mixed forests biome for all lifeforms except herb cover, which had the highest uncertainty in the tropical and subtropical grasslands, savannas, and shrublands biome.

B. Comparison With Other Cover Products

The comparison between our LPC models and the GFCC and NLCD products resulted in better agreement between our LPC predictions and the validation data for shrub, herb, and "other" cover, and lower performance of the LPC tree model, compared with the GFCC 2015 and the NLCD 2011 tree cover products (Table V).

The resulting analysis of the 2015 tree cover products [Fig. 4(a)] indicates some similarities between the GFCC and the LPC, specifically in the overprediction and underprediction pattern of both products, but a marked difference in the saturation of predicted values of the GFCC around 80% tree cover (as reported by [19]), which does not appear in the LPC, and higher overestimation around low cover values for the GFCC. The NLCD 2011 tree cover product [Fig. 4(b)] seems to consistently overestimate tree cover in low and intermediate cover values, which resulted in a higher MBE compared to LPC (NLCD: MBE = 17.8%; LPC: MBE = −8.5%). The NLCD product has a better agreement between prediction and validation values in high cover areas,

TABLE V
MODELS PERFORMANCE COMPARISON WITH OTHER COVER PRODUCTS

Model	RMSE (%)	MBE (%)	R	Samples
GFCC tree cover 2015	29.38	-19.11	0.80	1149717
LPC tree 2015	38.56	-28.46	0.68	
NLCD tree cover 2011	27.96	17.83	0.84	81218
LPC tree 2011	30.18	-8.55	0.76	
NLCD shrub 2016	19.53	12.01	0.42	1008684
LPC shrub 2016	9.24	3.95	0.55	
NLCD herb 2016	22.71	2.27	0.37	1025265
LPC herb 2016	20.13	1.26	0.52	
NLCD bare ground 2016	33.93	-19.56	0.29	1008480
LPC other 2016	24.8	-11.48	0.69	

which reflects in a lower RMSE and a higher correlation coefficient (NLCD: RMSE = 28%, $R = 0.8$; LPC: RMSE = 30.2%, $R = 0.76$). The analysis of the fractional components of the NLCD 2016 shrubland product [Fig. 4(c)–(e)] indicates a consistent overestimation of lifeform cover in low and intermediate cover values, while the LPC predictions show a better agreement with the validation data, which resulted in lower RMSE and MBE, and higher correlation values of

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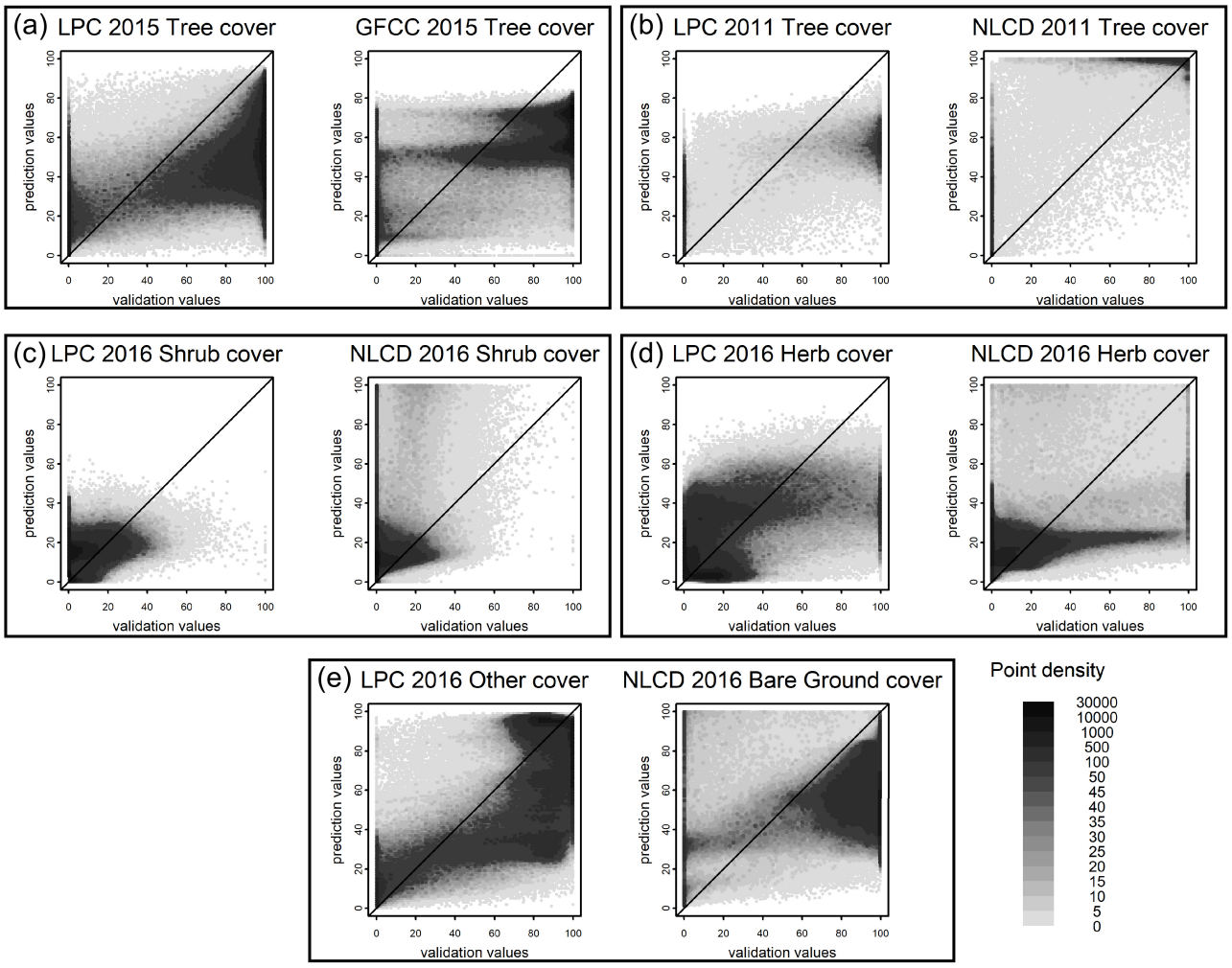


Fig. 4. Scatterplots of predicted cover versus observed cover values for comparison between the LPC product constructed in this study and the available continuous cover products. (a) LPC 2015 tree cover and GFCC 2015 tree cover product. (b) LPC 2011 tree cover and NLCD 2011 tree cover product. (c) LPC 2016 shrub cover and NLCD 2016 shrub cover product. (d) LPC 2016 herb cover and NLCD 2016 herb cover product. (e) LPC 2016 other cover and NLCD 2016 bare ground cover product. The 1:1 line is presented in black.

the LPC shrub, herb, and "other" predictions compared to the shrub, herb, and bare ground components of the NLCD product.

C. Qualitative Model Performance in Areas of Documented Lifeform Cover Change

The analysis of differences in percent cover indicates an overall correspondence between changes in predicted cover and the expected variations in lifeform cover for the selected land cover change processes.

In Fig. 5(a), we evaluated the differences in tree cover between 2014 and 2015 for an area in California to detect cover losses related to the King Fire [56]. The results show a clear area with tree cover reduction, which conforms to the King Fire's boundary and indicates tree cover loss of more than 40% in most of the fire-affected areas.

Fig. 5(b) shows a documented process of shrub encroachment in the Santa Rita Experimental Range (SRER) due to grazing and changes in the fire regime, which is believed to have initiated in the 1900s [57]. The difference in predicted cover between 1984 and 2018 shows a marked increase in shrub cover across the area of the SRER and its surroundings, with an increase of up to 40%.

Fig. 5(c) shows the differences in herb cover in the Monache meadow located in the southern Sierra Nevada range in California. This meadow experienced significant livestock grazing leading to changes in vegetation cover and bank erodibility [58]. The change cover analysis between 1984 and 2018 shows a reduction pattern in herb cover for most of the meadow areas, which could be related to an increase in floodplain erosion or to changes in vegetation cover from meadow riparian species to xeric upland species, as described in [58].

Fig. 5(d) shows the urbanization process in Detroit, MI, USA, between 1984 and 2018. The difference in the LPC maps shows a pattern of other cover increase in the periphery of the city, with changes in cover of over 45%, likely related to urban expansion into agricultural areas [59].

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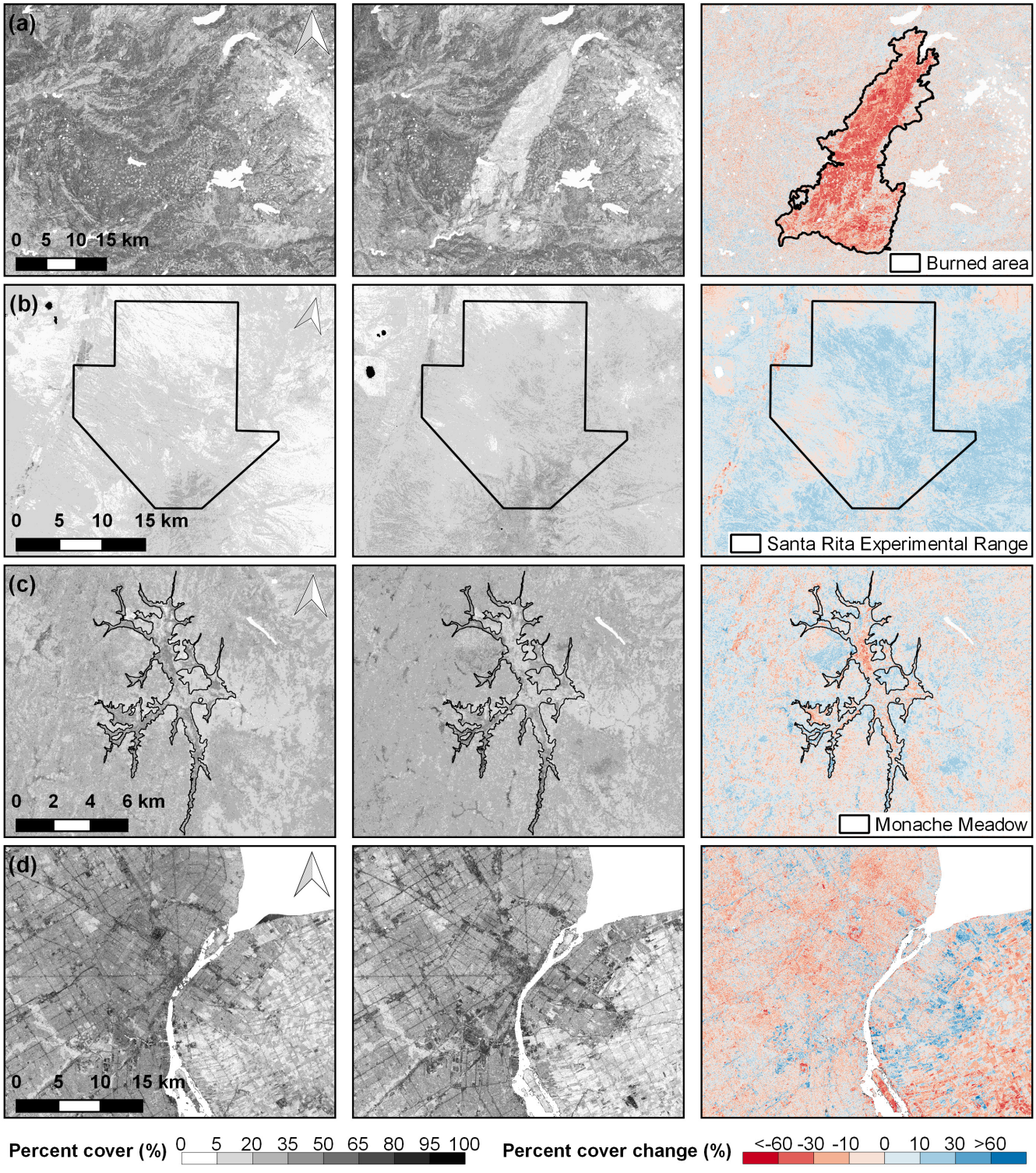


Fig. 5. Qualitative assessment of the LPC product using local examples of land cover change. (a) Differences in tree cover for the King Fire, CA, USA, between the years 2014 and 2015, centered at 120°29'31.2"W, 38°58'30"N. Burned area boundary from the Monitoring Trends in Burn Severity (http://www.mtbs.gov/; accessed July 20, 2018). (b) Shrub encroachment at the SRER, AZ, USA, between 1984 and 2018, centered at 110°51'30.24"W, 31°50'14.28"N. (c) Loss of Herb cover for the Monache meadow in CA, USA, between 1984 and 2018, centered at 118° 9'46.57"W, 36°11'6.56"N. (d) Urbanization process in Detroit, MI, USA, between 1984 and 2018, centered at 83°5'43.59"W, 42°18'16.49"N.

IV. Discussion

We produced four calibrated models of lifeform cover and their uncertainties for the conterminous USA, using airborne LiDAR/multispectral data and Landsat imagery. Our products capture the variability in percent cover of four lifeform classes for the conterminous USA and for the duration of most of the Landsat archive. These prediction maps represent a consistent dataset of medium spatial resolution (30 m) and long temporal

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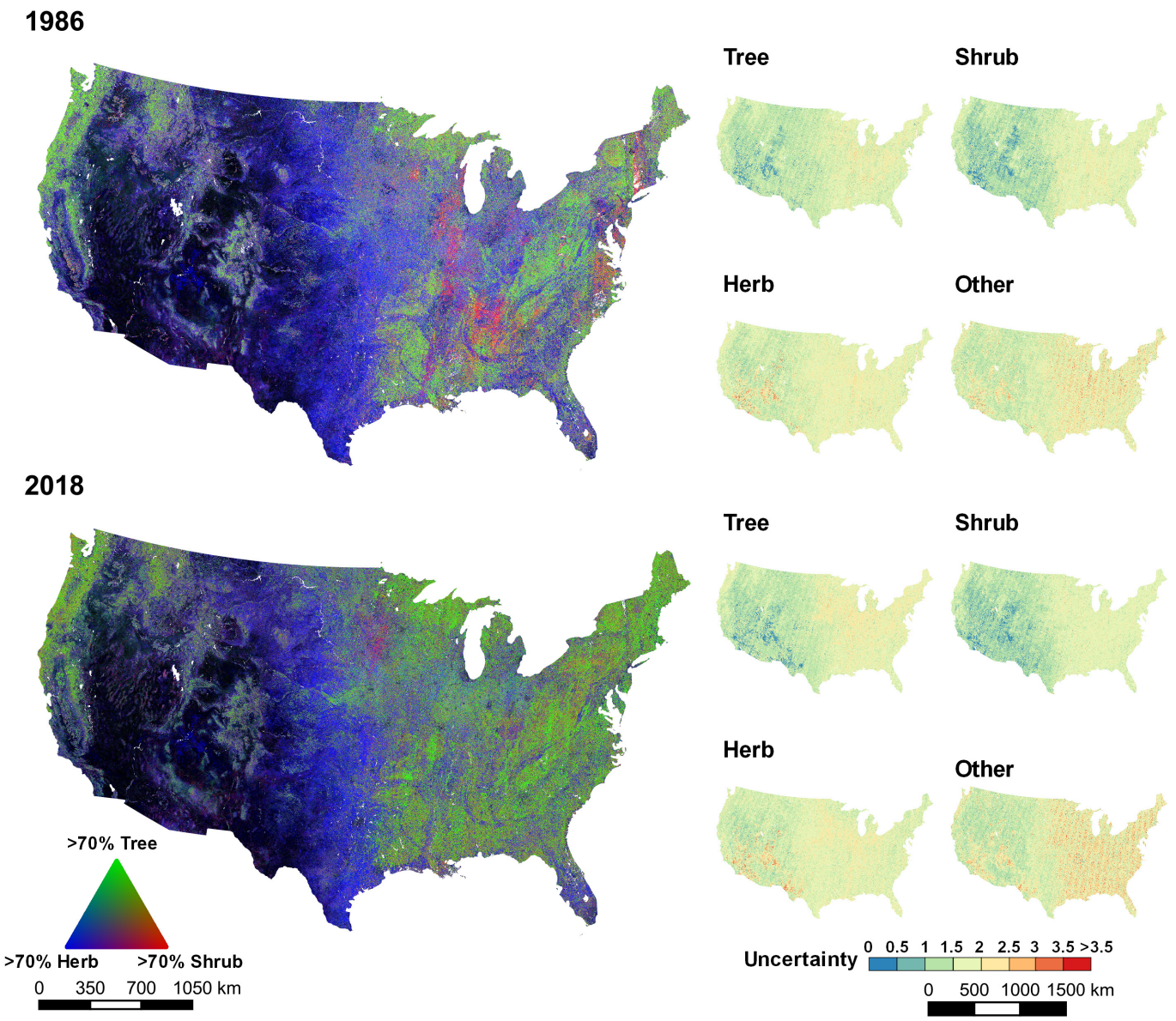


Fig. 6. Examples of LPC maps produced for the CONUS for the years 1986 and 2018 with the corresponding uncertainty map for each of the lifeform classes.

record that can facilitate the analysis of local and landscape level land cover change processes (Fig. 6). Moreover, this dataset can be a valuable input in the identification of drivers of land cover change, and the assessment of management and conservation strategies for different ecosystems in USA.

A. Performance of Lifeform Cover Models

The analysis of model performance shows good model calibration and similarities in performance statistics between the LPC models and other available continuous cover products. Correlation values against validation data for the tree lifeform ($R = 0.78$) are similar to those reported by the GFCC ($R = 0.81$) [19], and the shrub ($R = 0.41$), herb ($R = 0.73$), and "other" ($R = 0.89$) lifeforms have similar correlation values to those reported in the NLCD shrubland product (Shrub $R = 0.62$; Herbaceous $R = 0.74$; Bare Areas $R = 0.79$) [20]. We feel several factors contribute to the uncertainty in our estimates. First and foremost, there is almost certainly a limit to how spectrally separable even the broad lifeform classes are when using only six spectral bands from Landsat SR, which is corroborated by the most accurate cover estimates being the nonvegetation "other" class and by the prediction uncertainty analysis per biome. We found little evidence that the uncertainty per biome for each lifeform type was related to the amount of training data for each biome, which is not surprising given we used 15 000 training samples per biome-cover stratum. With that said, we do note that there is a lack of high shrub cover sites in our training dataset, which could explain a portion of the error in that particular cover class. Second, while we did have ample training data across the bulk of the U.S.'s biomes, we are lumping the entirety of the diversity of plant species and

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structural attributes under only three height-based categories. Highly spectrally unique examples of these classes certainly exist, which would contribute to our uncertainty.

Finally, near the extremes of cover estimation, part of the uncertainty in the predictions may be related to the bounded nature of the percent cover. Since percent cover cannot take values below 0% or above 100%, predicting mean values will inevitably result in overprediction and underprediction of lifeform cover at or near the bounded extremes of the distribution. The implementation of quantile regression models for mapping vegetation percent cover could also improve the prediction accuracy through the evaluation of prediction arguments other than the mean.

With all of this said, however, in the future, we can leverage the prediction uncertainty from our maps to help inform whether these errors may be solved by, e.g., adding new training data and identifying spectrally unique cover classes. In Fig. 7, we show an example of how our spatially explicit uncertainty can identify sites of low confidence, which can help us understand how to improve our models.

B. LPC Compared to Other Cover Products

The pattern of overestimation and underestimation of lifeform cover at the extreme values of the distribution that appears in the LPC models is also present in the continuous percent cover products included in the analysis [19], [52], and for the validation dataset of this study, the GFCC and the NLCD products show higher overestimation at low cover values compared to the LPC predictions. Considering that Sexton *et al.* [19] used ancillary data to improve the representation of low tree cover in their training dataset to reduce overestimation, the differences in prediction patterns between the GFCC and the LPC around low cover areas could be related to a better representation of different lifeform mixture conditions for low cover values of tree in the LPC model. Furthermore, the differences in the overestimation between the NLCD products and the LPC models could be associated with the rules used to identify lifeforms. While the LPC training data and the validation dataset in this study used height thresholds to separate lifeforms, the training data of the NLCD 2011 tree cover product were constructed with photointerpretation and tree lifeforms were identified regardless of their height [52]; the training dataset of the NLCD shrubland product was constructed using field information and the differentiation of fractional types was done by field identification without considering height [20]. Therefore, the overestimation of tree cover in the NLCD 2011 tree cover product could be the result of trees lower than 5 m that are interpreted as shrubs or herbs in the validation dataset, and the overestimation of fractional cover in the NLCD 2016 shrubland product could be due to misclassification of lifeform types in the validation dataset that do not correspond to the established height thresholds.

Regarding model construction, the LPC models differ from the GFCC and NLCD products in the types of predictor variables used in the regression models and the number of models constructed. In the LPC product, the only predictor variables are the Landsat SR V/NIR/SWIR bands and only four models were constructed for the conterminous USA, one for each lifeform. In comparison, the GFCC model

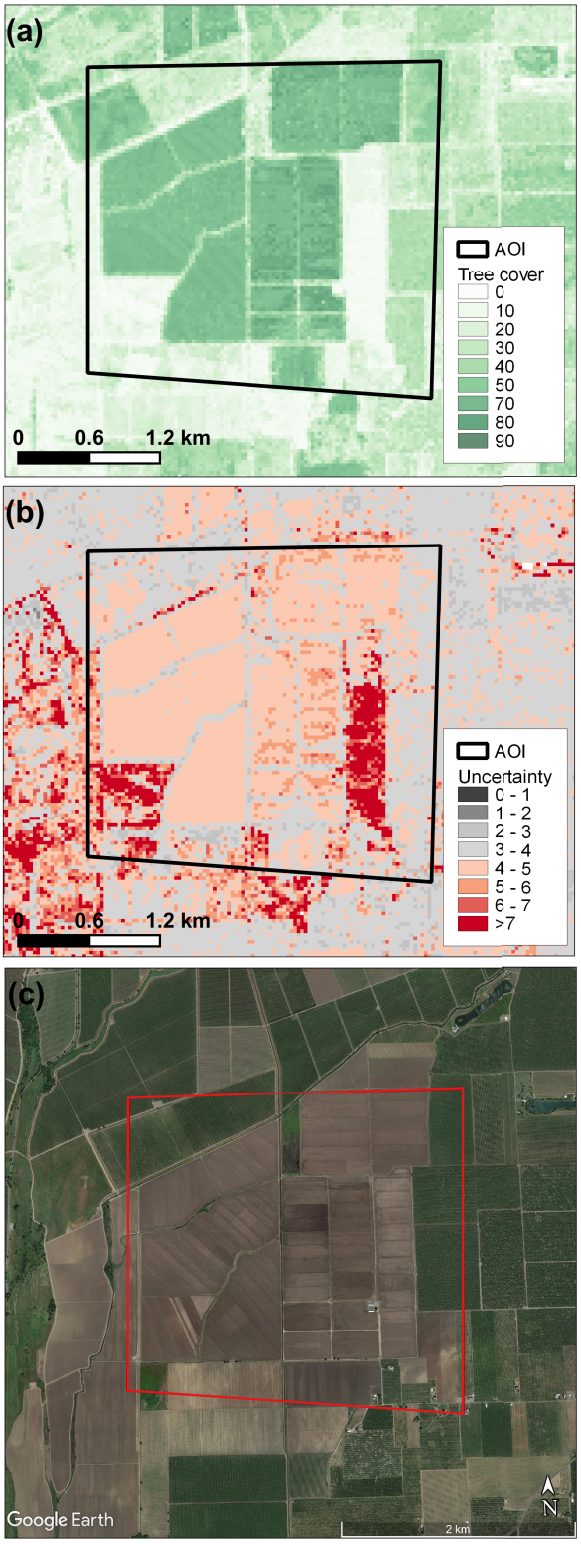


Fig. 7. Example of using the spatially explicit prediction uncertainty to identify and understand low-confidence sites. (a) This site's tree cover prediction shows a high tree cover, but (b) high uncertainty in these estimates. (c) High-resolution image shows that the high uncertainty estimates are correct, as this site is an agricultural field. Our LiDAR acquisitions have few agricultural sites within them, so this may be a candidate cover class to get more data to improve the modeling.

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includes SR and temperature as predictor variables, while the NLCD products include Landsat data and ancillary data, such as digital elevation models and previous cover products as explanatory variables. In both the GFCC and the NLCD products, specific models are fit for different areas or periods. For the GFCC product, regression tree algorithms were fit to each Landsat scene of the World Reference System 2 and for different epochs, and the NLCD products include regression models for each NLCD mapping zone or different climatic areas.

We decided on using a single model for each lifeform for the entire United States, rather than separate site-specific models, to take advantage of spectral similarities in vegetation across different areas of the country. Fitting separate regression models for specific locations and epochs would reduce the amount of variability of the predictor variables that need to be represented in the training data, reducing the size of the training dataset per model, in contrast to a single model for which a broad range of the spectral distribution needs to be represented in the training data to account for a larger variety of vegetation conditions. However, considering the large area of the CONUS, fitting several local models instead of a single global one would eventually result in overall larger training data requirements, as each local model would require its own training dataset. In addition, constructing separate site-specific models limits the extrapolation of the prediction maps for areas outside of the models' scope, resulting in seamlines along transition zones from, e.g., one biome classification to another.

C. Qualitative Assessment of LPC Maps

The analysis of local examples of land cover change using the LPC maps indicates that the product can track overall changes in lifeform cover in line with the patterns described in the literature. Moreover, the differences in lifeform cover for the CONUS between 1986 and 2018 shown in Fig. 6 correspond to regional vegetation change processes described in the literature. The recovery dynamics of the Northeastern forests due to changes in land use and abandonment of agricultural areas [60] appear as an increase in tree cover, paired with reductions of herb cover, for a large area of the eastern CONUS. The observed reduction in shrub cover for this same area could also be related to successional processes in which small trees are initially classified as shrub in 1986 and later as trees for the 2018 map as they reach the height threshold. Furthermore, studies of land cover change for similar periods show an increase in vegetation cover for the eastern CONUS. Song *et al.* [61] evaluated global changes in percent cover of tree canopy, short vegetation cover, and bare ground cover and found a 15% increase in tree canopy for USA, concentrated in the eastern part of the country. Chen *et al.* [62] evaluated global changes in annual average green leaf area from 2000 to 2017 and found that 33.3% of USA show greening trends (i.e., an increase in annual average green leaf area). Moreover, other studies, evaluating net primary productivity or NDVI, across similar periods agree on the existence of positive vegetation trends in USA that could be related to the increase in tree cover observed in Fig. 6 [63], [64], [65], [66].

Despite the overall agreement of the LPC products with local and regional land cover change processed described in the literature, it was beyond the scope of this analysis to evaluate the quantitative accuracy of the predicted LPC change values since independent sources of spatial information on vegetation cover change were not available.

Undoubtedly, further analysis of local and regional cases of land cover changes in which the LPC maps are evaluated against field data, and other sources of land cover information will refine the accuracy assessment of the LPC product, as well as identify areas of poor product performance for which additional training data might be necessary. We hope that future users of the LPC maps can offer feedback and detailed evaluations that facilitate the improvement of the product.

D. Possible Drawbacks of Using LiDAR/Multispectral Data

While using LiDAR/multispectral information as training data can vastly increase the sample size of a remote sensing analysis, resulting in significantly more of a landscape's heterogeneity being represented, some pitfalls related to the use of this approach must be considered. Foremost, the differences in data quality and resolution across different sources of information could represent an important source of error or of differences in accuracy. These data parameters will vary depending on the objectives of the studies for which the airborne data were acquired and the specifications of the equipment used in the observation platform. In particular, differences in point density and scan angle between different acquisitions can cause errors in the estimation of vegetation canopy [24] and should be considered when incorporating LiDAR/multispectral information into the construction of maps of vegetation. Second, LiDAR/multispectral data collection tends to be biased toward specific biomes or ecosystems of interest, resulting in a lack of availability of training and validation data for other areas with important representation of specific vegetation. In particular, degraded areas have been underrepresented in comparison to pristine areas, or ecosystems of higher conservational or economic interest [67]. Considering that data acquisition for the G-LiHT and NEON platforms follows specific research objectives, it is possible for vegetation with unique spectral characteristic to go unsampled.

V. Conclusion

The results of this study indicate that using airborne LiDAR/multispectral acquisitions as training and validation data for model construction can facilitate the inclusion of data from remote or difficult access areas into training and validation datasets. Furthermore, the efficient production of wall-to-wall training and validation data from airborne LiDAR/multispectral reduces the need to fit individual models for different areas, as the larger range of variation in the response and predictor variables across different sites are incorporated in the training data. This fusion of airborne data with Landsat information allows for the construction of moderate spatial and temporal resolution land cover maps that can be valuable inputs for evaluating ecological research questions at different scales and for linking mechanisms of

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land cover change at regional and global scales. A variety of land cover change processes, from subtle changes in fractional lifeform composition to rapid losses of vegetation cover, could be tracked and analyzed in various ecosystems across USA with annual maps of fractional land cover.

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